Antiderivatives and Riemann Sums

Section 1: Basic Antiderivatives & Power Rule for Integration

1 Evaluate the indefinite integral $\int (6x^2 - 4x + 7)dx$.

a. $2x^3 - 2x^2 + 7x + C$

b. $3x^3 - 2x^2 + 7x + C$

c. $12x - 4 + C$

d. $2x^3 - 4x^2 + 7x + C$

2 Find $\int (\frac{3}{x} + \frac{1}{x^2})dx$ for $x > 0$.

a. $3\ln x - \frac{1}{x} + C$

b. $3\ln x + \frac{1}{x} + C$

c. $-\frac{3}{x^2} - \frac{2}{x^3} + C$

d. $3\ln x - \frac{1}{2x} + C$

3 Find the general antiderivative of $f(x) = \sec^2 x + e^{3x}$.

a. $\tan x + \frac{1}{3}e^{3x} + C$

b. $\sec x \tan x + 3e^{3x} + C$

c. $\tan x + 3e^{3x} + C$

d. $2\sec x + \frac{1}{3}e^{3x} + C$

Section 2: Left and Right Riemann Sums

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4 Approximate $\int_0^4 x^2 dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14

b. 30

c. 21.33

d. 16

5 What is the Right Riemann sum approximation for the above integral with $n = 4$?

a. 30

b. 14

c. 22

d. 21.33

6 If $f(x)$ is strictly increasing on $[a, b]$, what is the relationship between Left and Right Riemann sums and the true integral value I?

a. Left Sum $< I <$ Right Sum

b. Right Sum $< I <$ Left Sum

c. Left Sum $=$ Right Sum

d. Cannot be determined without concavity

Section 3: Midpoint & Trapezoidal Approximations

7 Using the Trapezoidal Rule with $n = 4$ equal subintervals, approximate $\int_0^4 x^2 dx$.

a. 22

b. 21.33

c. 24

d. 20

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8 If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal Rule approximation compare to the exact integral?

a. Trapezoidal Rule overestimates the true value

b. Trapezoidal Rule underestimates the true value

c. Trapezoidal Rule is exact

d. Depends on whether f is increasing

9 Approximate $\int_0^2 x^3 dx$ using a Midpoint Riemann sum with $n = 2$ equal subintervals.

a. 3.5

b. 4.0

c. 4.5

d. 2.0

Section 4: Limit of a Riemann Sum & Definite Integrals

10 Which definite integral is equal to $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{2i}{n})^3 \frac{2}{n}$?

a. $\int_1^3 x^3 dx$

b. $\int_0^2 x^3 dx$

c. $\int_1^2 (1 + x)^3 dx$

d. $\int_0^1 (1 + 2x)^3 dx$

11 Evaluate $\int_1^3 x^3 dx$.

a. 20

b. 18

c. 26

d. 80

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12 Express $\int_0^5 \sqrt{x}\,dx$ as a limit of a Right Riemann sum.

a. $\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{\frac{5i}{n}}\frac{5}{n}$

b. $\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{\frac{i}{n}}\frac{5}{n}$

c. $\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{5i}\frac{1}{n}$

d. $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{5}{n}\sqrt{x_i}$

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